

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 2: Transmission Media & Noise

Course: Computer Networks (Credit Hours: 3)

Instructor: Muhammad Usman Ghani

1. WHAT IS TRANSMISSION MEDIA? (THE DATA ROAD)

Transmission Media is simply the path or bridge that carries data from device A to device B. If data is a car, transmission media is the road.

Type 1: Guided Media (Wired)

Data travels through a physical wire or cable.

- **Twisted Pair Cable:** Normal yellow/blue internet LAN cable. Wires are twisted together so they don't catch outside noise.
- **Coaxial Cable:** Thick TV cable with a single copper pin inside.
- **Fiber Optic Cable:** Thin glass pipe. Transmits data as **flashes of light**. Super fast!

Type 2: Unguided Media (Wireless)

Data travels through the air without any physical wire.

- **Radio Waves:** Wi-Fi and FM radio. Can go through house walls easily.
- **Microwaves:** Used by mobile phone towers and dishes. Only goes in a straight line.
- **Infrared:** TV remote control. Cannot go through walls (must point directly).

2. SIGNAL PROBLEMS (WHY SIGNALS GET WEAK OR DAMAGED)

When data travels a long distance, it does not stay perfect. Three bad things can happen:

1. Attenuation (Signal Getting Weak)

The signal loses power and fades away as it travels further.

Easy Example If you shout, people standing close hear you loudly, but people 300 meters away can hardly hear you.

How to fix it: Put a **Repeater** or **Booster** in the middle to make the signal strong again.

2. Distortion (Signal Loses Its Shape)

The signal changes its shape on the way. The receiver gets a broken or messy wave.

Easy Example Hearing someone speak underwater—sound reaches your ear, but the voice sounds weird and unclear.

3. Noise (Unwanted Outside Dirt / Interference)

Unwanted extra electricity mixes into our data and corrupts it.

Easy Example The static "shhh" sound on a phone call when there is thunder or rain outside.

3. FOUR COMMON TYPES OF NOISE

Noise Name	What is it? (In Simple Words)	Everyday Example
1. Thermal Noise	Normal heat inside copper wire causes electrons to shake. Cannot be stopped.	The light "hiss" sound coming from empty speakers.
2. Crosstalk	Two wires running side by side leak signals into each other.	Faintly hearing another person's call while on your phone.
3. Impulse Noise	A sudden, strong electric shock for a split second. Destroys data bits.	Turning on a heavy motor, fridge compressor, or lightning strikes.
4. Intermodulation	Two different signals share one cable and produce a third unwanted noise.	Two radio channels mixing together on the same frequency.

4. QUICK STUDENT QUESTIONS

Q1: Why does Fiber Optic cable have ZERO noise?

Ans: Because it sends **light** inside glass, not electricity. Electricity can catch noise, light does not!

Q2: Why does Wi-Fi become 1 bar in the bedroom far from the router?

Ans: Due to **Attenuation** (the Wi-Fi signal gets tired and weak over distance and walls).